OLLSCOIL NA hÉIREANN, CORCAIGH
THE NATIONAL UNIVERSITY OF IRELAND, CORK

COLÁISTE NA hOLLSCOILE, CORCAIGH
UNIVERSITY COLLEGE, CORK

Examination Session and Year	Autumn 2024
Module Code	EC4215
Module Title	Business Econometrics and Forecasting
Paper Number	1
External Examiner	Prof. Mac an Bhaird
The Head of the Department	Prof. Doran
Internal Examiners	Dr. Turner
Instructions to Candidates	(see below)
Duration of Paper	1.5 hours
Special Requirements	**Non-programmable calculators may be used**

Instructions to Candidates

Sections	No. of Questions to be attempted		% of total marks available
A	**Long Questions**	**2 out of 3**	**100%**

All questions within each section carry equal weight

Attempt the correct number of questions in each section: Only that number of questions will be graded. If you attempt more than the correct number, please indicate what questions you wish to have graded. If you fail to do this, the appropriate number of <u>lowest-numbered questions</u> of those you answered will be graded.

PLEASE DO NOT TURN THIS PAGE UNTIL INSTRUCTED TO DO SO
THEN ENSURE THAT YOU HAVE THE CORRECT EXAM PAPER

Section A: Long Questions (Answer 2 out of 3)

A1. **(a)** Suppose that a researcher is examining the earnings of a rock band over time and her data produces the following econometric regression results (standard errors in brackets):

$$E_t = 50.3 + \underset{(0.17)}{0.65}D_t + \underset{(1.8)}{5.2}T_t + \underset{(18.3)}{65.1}A_t \qquad \textit{(Eqn. 1.1)}$$

$\bar{R}^2 = 0.78$ n = 18

Where:
E_t = the earnings of the rock band in year t (in €000)
D_t = the number of song downloads for the band in year t (in thousands)
T_t = the number of tickets sold for concerts in year t (in thousands)
A_t = a dummy variable set to 1 if the band released a new album in year t, and 0 otherwise.

(i) Carefully interpret the above findings.

(35 per cent)

(ii) Does the above equation fit your expectations? Why or why not?

(15 per cent)

(iii) If you could add one more variable into the above regression, what would it be and why?

(10 per cent)

(b) Suppose that a researcher runs a regression to examine the factors that affect take-up of private health insurance in Ireland from Quarter 1, 2003 to Quarter 4, 2016 and gets the following results (standard errors in brackets):

$$T_t = 784.3469 + \underset{(0.008084)}{0.040072}C_t + \underset{(10.58059)}{9.218391}P_t - \underset{(0.707827)}{1.593816}I_t$$

$\bar{R}^2 = 0.56833$ n = 56
RSS = 76,986.76 ESS = 111,648.9

Where
T_t = Number of people with private health insurance in Quarter t (000s)
C_t = Personal consumption expenditure in Quarter t (€m, constant prices)
P_t = Labour force participation rate of those aged 15 and over in Quarter t (%)
I_t = Consumer price index for insurance related to health in Quarter t (Dec. 2016 = 100)

Question A1 continued on next page

Question A1 continued

(i) Looking at this output (without carrying out any statistical tests), would you expect the regression to be significant? Explain your answer.

(10 per cent)

(ii) Formally test this regression for significance at the 5% level of significance. Explain the steps you take.

(30 per cent)

A2. (a) Suppose a researcher has run a regression to examine the factors affecting cinema attendances in Ireland over a year and gets the following results (standard errors in brackets):

$$CT_t = 35.3 + \underset{(3.8)}{11.8}\, RF_t + \underset{(5.4)}{21.3}\, AD_t + \underset{(3.4)}{9.5}\, SH_t \qquad (\textit{Eqn. 2.1})$$

$N = 52$ Adjusted $R^2 = 0.582$

Where:

CT_t = the number of cinema tickets purchased in week t (000s)

RF_t = the average rainfall across Ireland in week t (in mm)

AD_t = the amount spent on cinema advertising in week t (€m)

SH_t = a dummy variable set to 1 if the week falls during the school holidays and 0 otherwise

Suppose the researcher is debating whether to include a variable to assess whether awards ceremonies (e.g. Oscars, Golden Globes, etc.) affect cinema attendances. If the researcher adds in this variable, she gets the following results:

$$CT_t = 36.8 + \underset{(3.7)}{12.6}\, RF_t + \underset{(4.9)}{18.4}\, AD_t + \underset{(3.3)}{9.3}\, SH_t + \underset{(2.1)}{9.2}\, AC_t \qquad (\textit{Eqn. 2.2})$$

$N = 52$ Adjusted $R^2 = 0.654$

Where:

AC_t = a dummy variable set to 1 if there is a major awards ceremony during the week and 0 otherwise

All of the other variables are as before.

Should the researcher include AC_t in her regression (i.e. go with Eqn. 2.2 instead of Eqn. 2.1)? Explain your answer.

(70 per cent)

Question A2 continued on next page

Question A2 continued

(b) Suppose that the 'true' regression model is:

Eqn. 2.1 $$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$$

Where u_i is the classical disturbance term.

However, suppose a researcher estimates the following regression:

Eqn. 2.2 $$Y_i = \alpha_0 + \alpha_1 X_{1i} + v_i$$

Assess the consequences of estimating the mis-specified model (Eqn. 2.2) instead of the 'true' model (Eqn. 2.1).

(30 per cent)

A3. **(a)** A researcher wanted to examine the determinants of expenditure on health across Europe, so she gathered data on the following indicators for a number of European countries:

HEXPTOT = Total health expenditure in 2017 (€m)
HEXPPUB = Public health expenditure in 2017 (€m)
GDP = Gross Domestic Product in 2017 (€m current prices)
POP = Total population on the 1st of January 2017 (number)
POP75PROP = Proportion of those aged 75+ in the population (%)
ALOS = Average length of inpatient hospital stay in 2017 (days)
NONSMOK = Proportion of non-smokers in the population in 2014 (%)
BEDS = Number of available beds in hospitals in 2017 (number)

Having considered the variables, the researcher decided to run a regression using HEXPTOT as the dependent variable and GDP, POP75PROP, ALOS and NONSMOK as the independent variables.

Question A3 continued on next page

Question A3 continued

She then ran a regression and obtained the following results:

Dependent Variable: HEXPTOT
Method: Least Squares
Date: 07/22/20 Time: 11:27
Sample (adjusted): 1 31
Included observations: 26 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	36413.90	48703.41	0.747666	0.4630
GDP	0.108243	0.003248	33.32736	0.0000
POP75PROP	-650.1453	1921.032	-0.338435	0.7384
ALOS	1534.125	3335.702	0.459911	0.6503
NONSMOK	-600.0428	627.6963	-0.955944	0.3500

R-squared	0.986665	Mean dependent var	56688.36
Adjusted R-squared	0.984126	S.D. dependent var	93740.11
S.E. of regression	11810.66	Akaike info criterion	21.76243
Sum squared resid	2.93E+09	Schwarz criterion	22.00438
Log likelihood	-277.9116	Hannan-Quinn criter.	21.83210
F-statistic	388.4654	Durbin-Watson stat	2.580435
Prob(F-statistic)	0.000000		

(i) Does anything in the above results suggest that this regression suffers from any econometric problems? Explain your answer.

(20 per cent)

(ii) The researcher then ran the White Test and obtained a Chi-squared statistic of 25.50482, compared with a critical Chi-squared statistic (for 14 degrees of freedom with a 5% level of significance) of 23.7. Does this result suggest the presence of heteroscedasticity? Explain your answer.

(20 per cent)

(b) Describe at least two different tests for detecting heteroscedasticity and outline the similarities and differences between them.

(60 per cent)

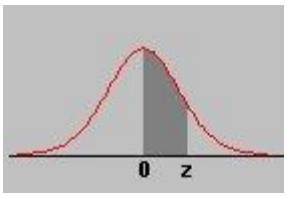
Appendix

A1: Selected Formulae

Slope coefficient: $\hat{\beta}_1 = \frac{\sum(X-\bar{X})(Y-\bar{Y})}{\sum(X-\bar{X})^2}$	Intercept coefficient: $\hat{\beta}_o = \bar{Y} - \hat{\beta}_1\bar{X}$
Standard Error of slope coefficient: $\text{se}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\sum x_i^2}}$	Standard Error of intercept coefficient: $\text{se}(\hat{\beta}_0) = \sqrt{\frac{\sum X_i^2}{n\sum x_i^2}}\hat{\sigma}$
Standard Error of estimated regression: $\hat{\sigma} = \sqrt{\frac{\sum e_i^2}{n-k-1)}}$	Total, Explained and Residual Sum of Squares: $\sum_{i=1}^{n}(Y_i-\bar{Y})^2 = \sum_{i=1}^{n}(\hat{Y}_i-\bar{Y})^2 + \sum_{i=1}^{n}{e_i}^2$
<u>Confidence Interval:</u> Confidence Interval = $\hat{\beta}_k \pm t_{crit} * \text{se}(\hat{\beta}_k)$	<u>t-test:</u> $t = \frac{\hat{\beta}_k - \beta_{\text{H0}}}{\text{se}(\hat{\beta}_k)} \sim t(n-k-1)\text{df}$
<u>F test (overall significance):</u> $F = \frac{\text{ESS/k}}{\text{RSS}/(n-k-1)} \sim \text{F}(\text{k,n}-k-1)$	<u>F test (overall significance):</u> $F = \frac{R^2/(k)}{(1-R^2)/(n-k-1)} \sim \text{F}(\text{k,n}-k-1)$
<u>F test (Linear Restrictions):</u> $F = \frac{(\text{RSS}_R - \text{RSS}_{\text{UR}})/\text{m}}{\text{RSS}_{\text{UR}}/(n-k-1)} \sim \text{F}(\text{m,n}-k-1)$	<u>F test (Linear Restrictions):</u> $F = \frac{(R_{\text{UR}}^2\text{-R}_R^2)/\text{m}}{(1\text{- R}_{\text{UR}}^2)/(n-k-1)} \sim \text{F}(\text{m,n}-k-1)$

A2: Areas Under Normal Curve

Area between 0 and z

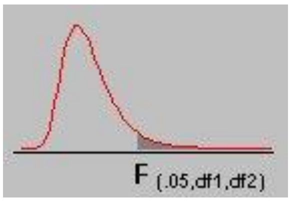


Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990

A3: Student t- Distribution

d.f.	Level of Significance for One-Tailed Test						
	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
	Level of Significance for Two-Tailed Test						
	0.2	0.1	0.05	0.02	0.01	0.002	0.001
1	3.078	6.314	12.706	31.821	63.656	318.289	636.578
2	1.886	2.920	4.303	6.965	9.925	22.328	31.600
3	1.638	2.353	3.182	4.541	5.841	10.214	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.894	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.689
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.660
30	1.310	1.697	2.042	2.457	2.750	3.385	3.646
32	1.309	1.694	2.037	2.449	2.738	3.365	3.622
34	1.307	1.691	2.032	2.441	2.728	3.348	3.601
36	1.306	1.688	2.028	2.434	2.719	3.333	3.582
38	1.304	1.686	2.024	2.429	2.712	3.319	3.566
40	1.303	1.684	2.021	2.423	2.704	3.307	3.551
50	1.299	1.676	2.009	2.403	2.678	3.261	3.496
75	1.293	1.665	1.992	2.377	2.643	3.202	3.425
100	1.290	1.660	1.984	2.364	2.626	3.174	3.390
∞	1.282	1.645	1.960	2.326	2.576	3.090	3.290

A4: Critical Values of the F Distribution at a 5 Percent Level of Significance



Degrees of Freedom for the Denominator	Degrees of Freedom for the Numerator																		
	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	inf
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01	249.05	250.10	251.14	252.20	253.25	254.31
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.37
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
inf	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

A5: Durbin-Watson d Statistic: Significance Points of dL and dU at 0.05 Level of Significance

k = numbers of parameters estimated in the model

	k = 2		k = 3		k = 4		k = 5		k = 6		k = 7	
n	dL	dU	dL	dU	dL	dU	dL	dU	dL	dU	dL	dU
6	0.610	1.400										
7	0.700	1.356	0.467	1.896								
8	0.763	1.332	0.559	1.777	0.367	2.287						
9	0.824	1.320	0.629	1.699	0.455	2.128	0.296	2.588				
10	0.879	1.320	0.697	1.641	0.525	2.016	0.376	2.414	0.243	2.822		
11	0.927	1.324	0.758	1.604	0.595	1.928	0.444	2.283	0.315	2.645	0.203	3.004
12	0.971	1.331	0.812	1.579	0.658	1.864	0.512	2.177	0.380	2.506	0.268	2.832
13	1.010	1.340	0.861	1.562	0.715	1.816	0.574	2.094	0.444	2.390	0.328	2.692
14	1.045	1.350	0.905	1.551	0.767	1.779	0.632	2.030	0.505	2.296	0.389	2.572
15	1.077	1.361	0.946	1.543	0.814	1.750	0.685	1.977	0.562	2.220	0.447	2.471
16	1.106	1.371	0.982	1.539	0.857	1.728	0.734	1.935	0.615	2.157	0.502	2.388
17	1.133	1.381	1.015	1.536	0.897	1.710	0.779	1.900	0.664	2.104	0.554	2.318
18	1.158	1.391	1.046	1.535	0.933	1.696	0.820	1.872	0.710	2.060	0.603	2.258
19	1.180	1.401	1.074	1.536	0.967	1.685	0.859	1.848	0.752	2.023	0.649	2.206
20	1.201	1.411	1.100	1.537	0.998	1.676	0.894	1.828	0.792	1.991	0.691	2.162
21	1.221	1.420	1.125	1.538	1.026	1.669	0.927	1.812	0.829	1.964	0.731	2.124
22	1.239	1.429	1.147	1.541	1.053	1.664	0.958	1.797	0.863	1.940	0.769	2.090
23	1.257	1.437	1.168	1.543	1.078	1.660	0.986	1.785	0.895	1.920	0.804	2.061
24	1.273	1.446	1.188	1.546	1.101	1.656	1.013	1.775	0.925	1.902	0.837	2.035
25	1.288	1.454	1.206	1.550	1.123	1.654	1.038	1.767	0.953	1.886	0.868	2.013
26	1.302	1.461	1.224	1.553	1.143	1.652	1.062	1.759	0.979	1.873	0.897	1.992
27	1.316	1.469	1.240	1.556	1.162	1.651	1.084	1.753	1.004	1.861	0.925	1.974
28	1.328	1.476	1.255	1.560	1.181	1.650	1.104	1.747	1.028	1.850	0.951	1.959
29	1.341	1.483	1.270	1.563	1.198	1.650	1.124	1.743	1.050	1.841	0.975	1.944
30	1.352	1.489	1.284	1.567	1.214	1.650	1.143	1.739	1.071	1.833	0.998	1.931
31	1.363	1.496	1.297	1.570	1.229	1.650	1.160	1.735	1.090	1.825	1.020	1.920
32	1.373	1.502	1.309	1.574	1.244	1.650	1.177	1.732	1.109	1.819	1.041	1.909
33	1.383	1.508	1.321	1.577	1.258	1.651	1.193	1.730	1.127	1.813	1.061	1.900
34	1.393	1.514	1.333	1.580	1.271	1.652	1.208	1.728	1.144	1.808	1.079	1.891
35	1.402	1.519	1.343	1.584	1.283	1.653	1.222	1.726	1.160	1.803	1.097	1.884
36	1.411	1.525	1.354	1.587	1.295	1.654	1.236	1.724	1.175	1.799	1.114	1.876
37	1.419	1.530	1.364	1.590	1.307	1.655	1.249	1.723	1.190	1.795	1.131	1.870
38	1.427	1.535	1.373	1.594	1.318	1.656	1.261	1.722	1.204	1.792	1.146	1.864
39	1.435	1.540	1.382	1.597	1.328	1.658	1.273	1.722	1.218	1.789	1.161	1.859
40	1.442	1.544	1.391	1.600	1.338	1.659	1.285	1.721	1.230	1.786	1.175	1.854
41	1.449	1.549	1.399	1.603	1.348	1.660	1.296	1.720	1.243	1.784	1.189	1.849
42	1.456	1.553	1.407	1.606	1.357	1.662	1.306	1.720	1.255	1.781	1.202	1.845
43	1.463	1.558	1.415	1.609	1.366	1.663	1.317	1.720	1.266	1.779	1.215	1.841
44	1.469	1.562	1.423	1.612	1.375	1.665	1.326	1.720	1.277	1.778	1.227	1.838
45	1.475	1.566	1.430	1.615	1.383	1.666	1.336	1.720	1.287	1.776	1.238	1.835
46	1.481	1.570	1.437	1.618	1.391	1.668	1.345	1.720	1.298	1.775	1.250	1.832
47	1.487	1.574	1.444	1.620	1.399	1.669	1.354	1.720	1.307	1.774	1.260	1.829
48	1.493	1.578	1.450	1.623	1.406	1.671	1.362	1.721	1.317	1.773	1.271	1.826
49	1.498	1.581	1.456	1.626	1.414	1.672	1.370	1.721	1.326	1.772	1.281	1.824
50	1.503	1.585	1.462	1.628	1.421	1.674	1.378	1.721	1.335	1.771	1.291	1.822
51	1.509	1.588	1.468	1.631	1.427	1.675	1.386	1.722	1.343	1.770	1.300	1.820
52	1.514	1.592	1.474	1.633	1.434	1.677	1.393	1.722	1.351	1.769	1.309	1.818
53	1.518	1.595	1.480	1.636	1.440	1.678	1.400	1.723	1.359	1.769	1.318	1.817
54	1.523	1.598	1.485	1.638	1.446	1.680	1.407	1.723	1.367	1.768	1.326	1.815
55	1.528	1.601	1.490	1.641	1.452	1.681	1.414	1.724	1.374	1.768	1.334	1.814
56	1.532	1.605	1.495	1.643	1.458	1.683	1.420	1.725	1.382	1.768	1.342	1.812
57	1.536	1.608	1.500	1.645	1.464	1.684	1.426	1.725	1.389	1.768	1.350	1.811
58	1.540	1.610	1.505	1.647	1.469	1.686	1.433	1.726	1.395	1.767	1.358	1.810
59	1.545	1.613	1.510	1.650	1.474	1.687	1.438	1.727	1.402	1.767	1.365	1.809
60	1.549	1.616	1.514	1.652	1.480	1.689	1.444	1.727	1.408	1.767	1.372	1.808
61	1.552	1.619	1.519	1.654	1.485	1.690	1.450	1.728	1.415	1.767	1.379	1.807
62	1.556	1.622	1.523	1.656	1.490	1.692	1.455	1.729	1.421	1.767	1.385	1.807
63	1.560	1.624	1.527	1.658	1.494	1.693	1.461	1.730	1.427	1.767	1.392	1.806
64	1.563	1.627	1.532	1.660	1.499	1.695	1.466	1.730	1.432	1.767	1.398	1.805
65	1.567	1.629	1.536	1.662	1.503	1.696	1.471	1.731	1.438	1.767	1.404	1.805
66	1.570	1.632	1.539	1.664	1.508	1.697	1.476	1.732	1.443	1.767	1.410	1.804
67	1.574	1.634	1.543	1.666	1.512	1.699	1.481	1.733	1.449	1.768	1.416	1.804
68	1.577	1.637	1.547	1.668	1.516	1.700	1.485	1.733	1.454	1.768	1.422	1.803
69	1.580	1.639	1.551	1.670	1.521	1.701	1.490	1.734	1.459	1.768	1.427	1.803
70	1.583	1.641	1.554	1.672	1.525	1.703	1.494	1.735	1.464	1.768	1.433	1.802
75	1.598	1.652	1.571	1.680	1.543	1.709	1.515	1.739	1.487	1.770	1.458	1.801
80	1.611	1.662	1.586	1.688	1.560	1.715	1.534	1.743	1.507	1.772	1.480	1.801
85	1.624	1.671	1.600	1.696	1.575	1.721	1.550	1.747	1.525	1.774	1.500	1.801
90	1.635	1.679	1.612	1.703	1.589	1.726	1.566	1.751	1.542	1.776	1.518	1.801
95	1.645	1.687	1.623	1.709	1.602	1.732	1.579	1.755	1.557	1.778	1.535	1.802
100	1.654	1.694	1.634	1.715	1.613	1.736	1.592	1.758	1.571	1.780	1.550	1.803
200	1.758	1.778	1.748	1.789	1.738	1.799	1.728	1.810	1.718	1.820	1.707	1.831

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